Neural Layer Matrices & Activations

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

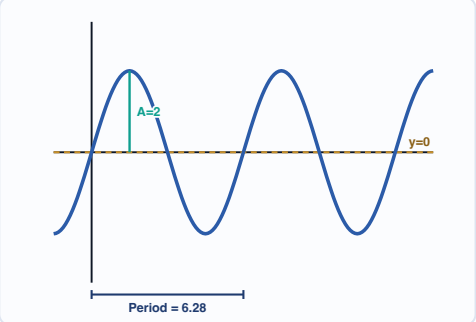
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 D	Q4 D	Q5 C
Q6 C	Q7 A	Q8 A	Q9 B	Q10 D
Q11 C	Q12 C	Q13 D	Q14 C	Q15 D
Q16 C	Q17 B	Q18 D	Q19 D	Q20 B
Q21 A	Q22 C	Q23 D	Q24 C	Q25 B

Neural Layer Matrices & Activations Practice Exercises

1 MCQ

A fully-connected dense layer maps an input $\mathbf{x} \in \mathrm{R}^{50}$ to $\mathbf{y} \in \mathrm{R}^{25}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?



- **A** 1275 ✓
- **B** 1250
- **C** 75
- **D** 1325

WORKED SOLUTION & VERIFICATION

The weight matrix $\mathbf{W}$ has shape 25×50, giving 1250 weights. The bias vector $\mathbf{b}$ has 25 parameters. Total = $1250 + 25 = 1275$.

2 MCQ

A fully-connected dense layer maps an input $\mathbf{x} \in \mathrm{R}^{60}$ to $\mathbf{y} \in \mathrm{R}^{30}$ using $\mathbf{y} = \mathbf{W}\mathbf{x} + \mathbf{b}$. How many total learnable parameters are in this layer?

- **A** 1830 ✓
- **B** 1800
- **C** 90
- **D** 1890

WORKED SOLUTION & VERIFICATION

The weight matrix $\mathbf{W}$ has shape 30×60, giving 1800 weights. The bias vector $\mathbf{b}$ has 30 parameters. Total = $1800 + 30 = 1830$.

3 MCQ

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -4$ and $z_2 = 6$.

A $f(z_1) = 4,\ f(z_2) = 6$

B $f(z_1) = -4,\ f(z_2) = 6$

C $f(z_1) = 0,\ f(z_2) = 0$

D $f(z_1) = 0,\ f(z_2) = 6$ ✓

WORKED SOLUTION & VERIFICATION

By definition of the Rectified Linear Unit: $\text{ReLU}(z) = z$ if $z > 0$, and 0 if $z \leq 0$. Thus $f(-4) = 0$ and $f(6) = 6$.

4 MCQ

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -6$ and $z_2 = 9$.

A $f(z_1) = 6,\ f(z_2) = 9$

B $f(z_1) = -6,\ f(z_2) = 9$

C $f(z_1) = 0,\ f(z_2) = 0$

D $f(z_1) = 0,\ f(z_2) = 9$ ✓

WORKED SOLUTION & VERIFICATION

By definition of the Rectified Linear Unit: $\text{ReLU}(z) = z$ if $z > 0$, and 0 if $z \leq 0$. Thus $f(-6) = 0$ and $f(9) = 9$.

5 MCQ

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -10$ and $z_2 = 15$.

A $f(z_1) = 0,\ f(z_2) = 0$

B $f(z_1) = -10,\ f(z_2) = 15$

C $f(z_1) = 0,\ f(z_2) = 15$ ✓

D $f(z_1) = 10,\ f(z_2) = 15$

WORKED SOLUTION & VERIFICATION

By definition of the Rectified Linear Unit: $\text{ReLU}(z) = z$ if $z > 0$, and 0 if $z \leq 0$. Thus $f(-10) = 0$ and $f(15) = 15$.

6 MCQ

Evaluate the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -12$ and $z_2 = 18$.

A $f(z_1) = 0,\ f(z_2) = 0$

B $f(z_1) = -12,\ f(z_2) = 18$

C $f(z_1) = 0,\ f(z_2) = 18$ ✓

D $f(z_1) = 12,\ f(z_2) = 18$

WORKED SOLUTION & VERIFICATION

By definition of the Rectified Linear Unit: $\text{ReLU}(z) = z$ if $z > 0$, and 0 if $z \leq 0$. Thus $f(-12) = 0$ and $f(18) = 18$.

7 MCQ

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.20$, what is the value of the derivative $\sigma'(z_0)$?

- **A** 0.1600 ✓
- **B** 0.4000
- **C** 0.8000
- **D** 0.0400

WORKED SOLUTION & VERIFICATION

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.20$: $\sigma'(z_0) = (0.20)(1 - 0.20) = (0.20)(0.80) = 0.1600$.

8 MCQ

The sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.40$, what is the value of the derivative $\sigma'(z_0)$?

- **A** 0.2400 ✓
- **B** 0.8000
- **C** 0.6000
- **D** 0.1600

WORKED SOLUTION & VERIFICATION

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.40$: $\sigma'(z_0) = (0.40)(1 - 0.40) = (0.40)(0.60) = 0.2400$.

9 MCQ

Determine the numerical value of the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -6$ and $z_2 = 9$.

- **A** $f(z_1) = 0,\ f(z_2) = 0$
- **B** $f(z_1) = 0,\ f(z_2) = 9$ ✓
- **C** $f(z_1) = 6,\ f(z_2) = 9$
- **D** $f(z_1) = -6,\ f(z_2) = 9$

WORKED SOLUTION & VERIFICATION

By definition of the Rectified Linear Unit: $\text{ReLU}(z) = z$ if $z > 0$, and 0 if $z \leq 0$. Thus $f(-6) = 0$ and $f(9) = 9$.

10 MCQ

Calculate the simplified result of the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -10$ and $z_2 = 15$.

- **A** $f(z_1) = 0,\ f(z_2) = 0$
- **B** $f(z_1) = -10,\ f(z_2) = 15$
- **C** $f(z_1) = 10,\ f(z_2) = 15$
- **D** $f(z_1) = 0,\ f(z_2) = 15$ ✓

WORKED SOLUTION & VERIFICATION

By definition of the Rectified Linear Unit: $\text{ReLU}(z) = z$ if $z > 0$, and 0 if $z \leq 0$. Thus $f(-10) = 0$ and $f(15) = 15$.

11 MCQ

Find the exact value of the expression the output of the standard ReLU activation function $f(z) = \max(0, z)$ for inputs $z_1 = -12$ and $z_2 = 18$.

A $f(z_1) = 0,\ f(z_2) = 0$

B $f(z_1) = -12,\ f(z_2) = 18$

C $f(z_1) = 0,\ f(z_2) = 18$ ✓

D $f(z_1) = 12,\ f(z_2) = 18$

WORKED SOLUTION & VERIFICATION

By definition of the Rectified Linear Unit: $\text{ReLU}(z) = z$ if $z > 0$, and 0 if $z \leq 0$. Thus $f(-12) = 0$ and $f(18) = 18$.

12 MCQ

In this problem, the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.40$, what is the value of the derivative $\sigma'(z_0)$?

A 0.1600

B 0.8000

C 0.2400 ✓

D 0.6000

WORKED SOLUTION & VERIFICATION

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.40$: $\sigma'(z_0) = (0.40)(1 - 0.40) = (0.40)(0.60) = 0.2400$.

13 MCQ

Observe that the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.60$, what is the value of the derivative $\sigma'(z_0)$?

A 0.4000

B 0.3600

C 1.2000

D 0.2400 ✓

WORKED SOLUTION & VERIFICATION

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.60$: $\sigma'(z_0) = (0.60)(1 - 0.60) = (0.60)(0.40) = 0.2400$.

14 MCQ

Suppose the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 0.80$, what is the value of the derivative $\sigma'(z_0)$?

A 1.6000

B 0.2000

C 0.1600 ✓

D 0.6400

WORKED SOLUTION & VERIFICATION

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Substituting $\sigma(z_0) = 0.80$: $\sigma'(z_0) = (0.80)(1 - 0.80) = (0.80)(0.20) = 0.1600$.

15 MCQ

Notice that the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 1.00$, what is the value of the derivative $\sigma'(z_0)$?

A 2.0000

B 1.0000

C 13

D 0.0000 ✓

WORKED SOLUTION & VERIFICATION

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1-\sigma(z))$. Substituting $\sigma(z_0) = 1.00$: $\sigma'(z_0) = (1.00)(1-1.00) = (1.00)(0.00) = 0.0000$.

16 MCQ

Consider the sigmoid activation function is $\sigma(z) = \frac{1}{1+e^{-z}}$. If $\sigma(z_0) = 1.20$, what is the value of the derivative $\sigma'(z_0)$?

A -0.2000

B 1.4400

C -0.2400 ✓

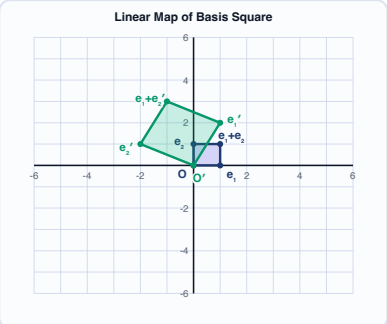
D 2.4000

WORKED SOLUTION & VERIFICATION

The derivative of sigmoid satisfies $\sigma'(z) = \sigma(z)(1-\sigma(z))$. Substituting $\sigma(z_0) = 1.20$: $\sigma'(z_0) = (1.20)(1-1.20) = (1.20)(-0.20) = -0.2400$.

17 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 9 \\ 1 \end{bmatrix}$

B $\begin{bmatrix} 0 \\ 9 \end{bmatrix}$ ✓

C $\begin{bmatrix} -5 \\ 9 \end{bmatrix}$

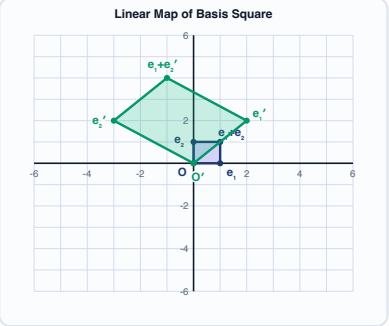
D $\begin{bmatrix} 5 \\ 9 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Pre-activation $\mathbf{z} = W\mathbf{x} + \mathbf{b} = [-3, 9]^\top$. Applying elementwise $\max(0, z_i)$ yields $[0, 9]^\top$.

18 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 2 & -3 \\ 2 & 2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 2 \\ 12 \end{bmatrix}$

B $\begin{bmatrix} 12 \\ 1 \end{bmatrix}$

C $\begin{bmatrix} -2 \\ 12 \end{bmatrix}$

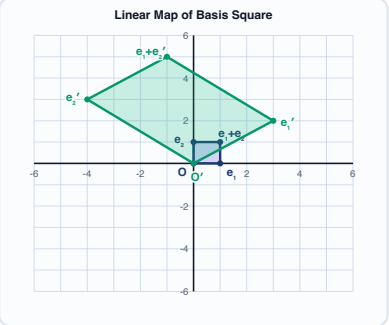
D $\begin{bmatrix} 0 \\ 12 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

Pre-activation $\mathbf{z} = W\mathbf{x} + \mathbf{b} = [0, 12]^\top$. Applying elementwise $\max(0, z_i)$ yields $[0, 12]^\top$.

19 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 3 & -4 \\ 2 & 3 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 15 \\ 6 \end{bmatrix}$

B $\begin{bmatrix} 3 \\ 15 \end{bmatrix}$

C $\begin{bmatrix} 7 \\ 15 \end{bmatrix}$

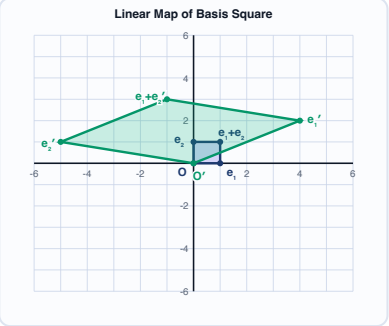
D $\begin{bmatrix} 5 \\ 15 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

Pre-activation $\mathbf{z} = W\mathbf{x} + \mathbf{b} = [5, 15]^\top$. Applying elementwise $\max(0, z_i)$ yields $[5, 15]^\top$.

20 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 4 & -5 \\ 2 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 10 \\ 15 \end{bmatrix}$

B $\begin{bmatrix} 12 \\ 15 \end{bmatrix}$ ✓

C $\begin{bmatrix} 15 \\ 13 \end{bmatrix}$

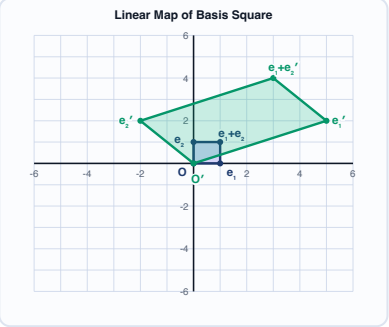
D $\begin{bmatrix} 14 \\ 15 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Pre-activation $\mathbf{z} = W\mathbf{x} + \mathbf{b} = [12, 15]^\top$. Applying elementwise $\max(0, z_i)$ yields $[12, 15]^\top$.

21 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 5 & -2 \\ 2 & 2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 25 \\ 18 \end{bmatrix}$ ✓

B $\begin{bmatrix} 23 \\ 18 \end{bmatrix}$

C $\begin{bmatrix} 18 \\ 26 \end{bmatrix}$

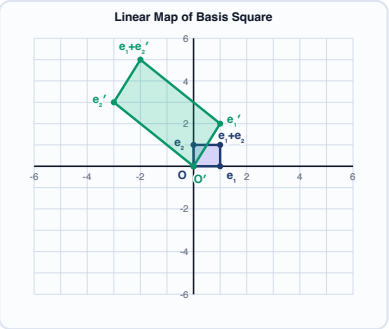
D $\begin{bmatrix} 27 \\ 18 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Pre-activation $\mathbf{z} = W\mathbf{x} + \mathbf{b} = [25, 18]^\top$. Applying elementwise $\max(0, z_i)$ yields $[25, 18]^\top$.

22 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 1 & -3 \\ 2 & 3 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} -1 \\ 21 \end{bmatrix}$

B $\begin{bmatrix} 3 \\ 21 \end{bmatrix}$

C $\begin{bmatrix} 1 \\ 21 \end{bmatrix}$ ✓

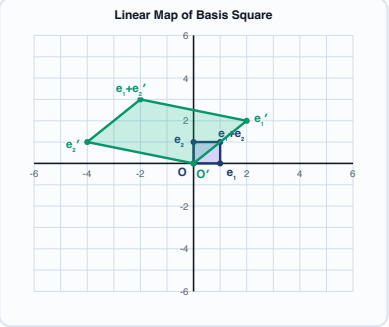
D $\begin{bmatrix} 21 \\ 2 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Pre-activation $\mathbf{z} = W\mathbf{x} + \mathbf{b} = [1, 21]^\top$. Applying elementwise $\max(0, z_i)$ yields $[1, 21]^\top$.

23 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 2 & -4 \\ 2 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} -9 \\ 10 \end{bmatrix}$

B $\begin{bmatrix} 9 \\ 10 \end{bmatrix}$

C $\begin{bmatrix} 10 \\ 1 \end{bmatrix}$

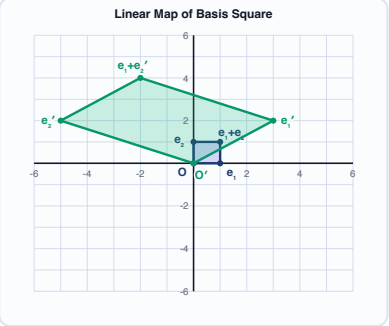
D $\begin{bmatrix} 0 \\ 10 \end{bmatrix}$ ✓

WORKED SOLUTION & VERIFICATION

Pre-activation $\mathbf{z} = W\mathbf{x} + \mathbf{b} = [-7, 10]^\top$. Applying elementwise $\max(0, z_i)$ yields $[0, 10]^\top$.

24 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 3 & -5 \\ 2 & 2 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 14 \\ 1 \end{bmatrix}$

B $\begin{bmatrix} -6 \\ 14 \end{bmatrix}$

C $\begin{bmatrix} 0 \\ 14 \end{bmatrix}$ ✓

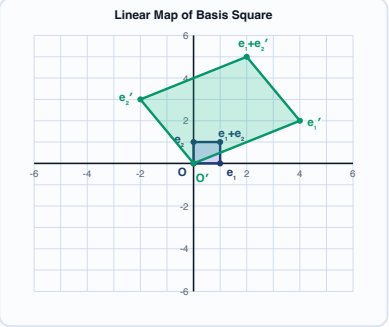
D $\begin{bmatrix} 6 \\ 14 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Pre-activation $\mathbf{z} = W\mathbf{x} + \mathbf{b} = [-4, 14]^\top$. Applying elementwise $\max(0, z_i)$ yields $[0, 14]^\top$.

25 MCQ

A dense neural layer computes $\mathbf{a} = \mathrm{ReLU}(W\mathbf{x} + \mathbf{b})$ where $\mathrm{ReLU}(z) = \max(0, z)$, $W = \begin{bmatrix} 4 & -2 \\ 2 & 3 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, and $\mathbf{b} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$. Find $\mathbf{a}$.



A $\begin{bmatrix} 11 \\ 18 \end{bmatrix}$

B $\begin{bmatrix} 9 \\ 18 \end{bmatrix}$ ✓

C $\begin{bmatrix} 7 \\ 18 \end{bmatrix}$

D $\begin{bmatrix} 18 \\ 10 \end{bmatrix}$

WORKED SOLUTION & VERIFICATION

Pre-activation $\mathbf{z} = W\mathbf{x} + \mathbf{b} = [9, 18]^\top$. Applying elementwise $\max(0, z_i)$ yields $[9, 18]^\top$.